

TradeAlerts Pro Suite

> **Built by Ben McCall, powered by Max the AI Sidekick**

■ Overview

TradeAlerts is a next-level, pro-grade stock scanning, simulation, and backtesting platform designed for both rapid signals and in-depth strategy analysis. Featuring a clean black UI, live dashboard, automated simulation, backtest engine, and integrated news failover, it's built for real traders, by a real trader.

Features

- **Entry/Exit Strategies:** Single-Entry Only, Trailing Stops, Sell-After-Days
- **Capital Allocation:** Starting Cash and Max Per Trade controls for both simulation and backtest
- **Pure black theme** with modern nav bar and responsive layout
- **Alerts Dashboard:** Real-time signal display, Prime filtering, local time, non-repeating cleared alerts
- **Simulation Dashboard:** Tracks holdings, cash, unrealized P&L, and logs every simulated trade
- **Backtest Engine:** Run 1-year strategy backtests, export results, and compare configs
- **Live Config Editing:** Tweak all scanner/backtest variables from the UI
- **Redundant News Feeds:** Multiple API keys auto-rotated, failover for news limits
- **SQLite Backend:** Each major feature (alerts, simulation, backtest) uses its own DB for reliability
- **Engineer-Friendly Controls:** "wr mem" (save), "show run" (status), and easy file structure

File Structure

```
/TradeAlerts/  
dashboard.py  
simulation_service.py  
backtest_service.py  
alert_service.py  
config_live.json  
config_backtest.json  
simulation.db  
backtest.db  
alerts.db  
templates/  
alerts.html  
simulation.html  
backtest.html  
static/  
style.css  
init_alerts_db.sql  
init_simulation_db.sql  
init_backtest_db.sql  
README.md
```

yaml

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Quick Start

1. ****Install Dependencies****:

pip install flask pytz

markdown

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2. ****Initialize Databases**** (see instructions above)

3. ****Run the App****:

python dashboard.py

yaml

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Then open `http://localhost:5000` in your browser.

Usage Tips

- Edit live or backtest configs from the Settings page.
- Use “Clear” buttons to dismiss alerts (they won’t return until a **true** new signal appears).
- Backtest engine logs all trade results—downloadable for analysis.
- News quotas are managed automatically; add API keys in the config as needed.

Credits

- ****Ben McCall**** – Lead developer, system design, and inspiration
- ****Max (AI Sidekick)**** – Assistant, code generation, documentation, and future-proofing

For questions, tweaks, or new feature requests, just ask Max!

Full Scanner Simulation

Our full scanner simulation engine iterates through the entire symbol universe, applies all enabled indicator filters (SMA, RSI, MACD histogram, Bollinger Bands breakout, VWAP threshold, etc.), and simulates trades in real-time or historical replay mode. It logs buy/sell events, reasons for decisions, and portfolio performance over time.

Full Scanner Backtesting

The backtesting module replays historical 1-minute and 1-day bar data across the full symbol set, applying the same indicator logic and trade execution rules. It calculates detailed metrics including win rate, average return, maximum drawdown, and Sharpe ratio—enabling side-by-side comparison of strategy variations.

Complex Indicator Logic

We support a full suite of technical indicators:

- ****Simple Moving Average (SMA)****: Configurable lookback and cross-over triggers.
- ****Relative Strength Index (RSI)****: Overbought/oversold thresholds and slope detection.

- **MACD Histogram:** Fast/slow EMA spans and signal line for momentum.
- **Bollinger Bands:** Customizable length and standard-deviation multipliers for breakout detection.
- **Volume Weighted Average Price (VWAP):** Intraday benchmark with threshold offsets.
- **Custom Combinations:** Flexible toggles for mixing any subset of the above filters in any order.